

August 24, 2026

MIR-26-26

Contact of Bulk Carrier *Belle Mer* Anchor with Subsea Cable

On November 15, 2024, about 1211 local time, the cargo vessel *Belle Mer* was anchored in the Tongass Narrows, near Ketchikan, Alaska, when the ship dragged anchor and damaged a subsea fiber optic cable (see figure 1 and figure 2).¹ There were no injuries, and no pollution was reported. There was no damage to the vessel. Damage to the subsea cable was estimated at \$240,000.

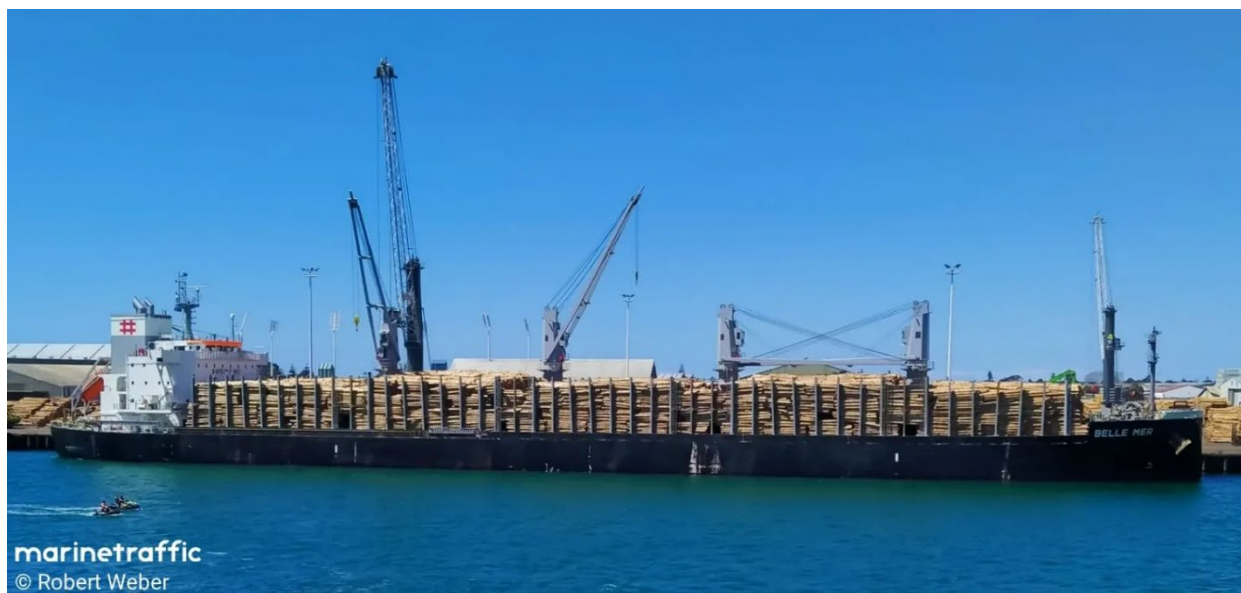


Figure 1. *Belle Mer* moored in October 2023. (Source: Robert Weber, marinetraffic.com)

¹ (a) In this report, all times are Alaska standard time, and all miles are nautical miles (1.15 statute miles) unless otherwise noted. (b) Visit [nts.gov](https://www.nts.gov) to find additional information in the [public docket](#) for this NTSB investigation (case no. DCA25FM005). Use the [CAROL Query](#) to search investigations.

Casualty Summary

NTSB casualty category	Other
Location	Tongass Narrows, near Ketchikan, Alaska 55°24.82' N, 131°47.07' W
Date	November 15, 2024
Time	1211 Alaska standard time (coordinated universal time –9 hrs)
Persons on board	21
Injuries	None
Property damage	\$240,000
Environmental damage	None
Weather	Visibility 10 mi, clear, calm winds, air temperature 45°F, water temperature 48°F
Waterway information	Narrows; width 7,300 ft, depth 242 ft, slack current

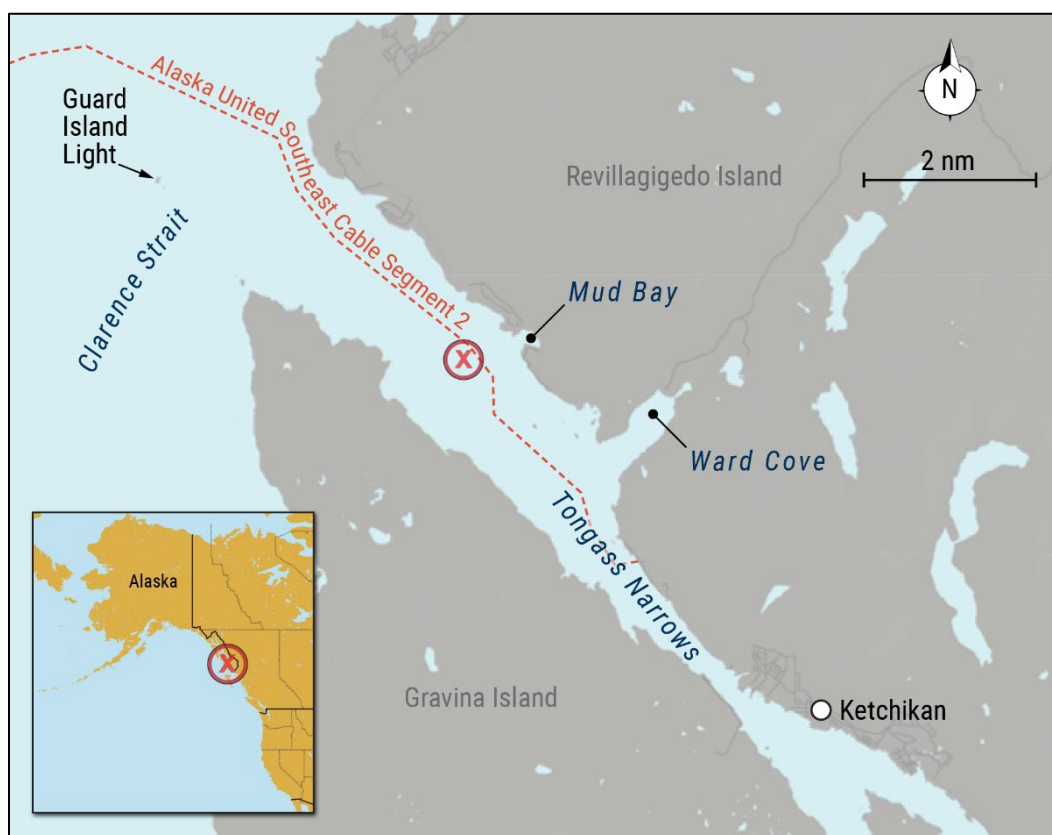


Figure 2. Area where the *Belle Mer* anchor struck a subsea cable, as indicated by a circled X. (Background sources: Google Maps, National Oceanic and Atmospheric Administration [NOAA] Electronic Navigational Chart [ENC] Viewer, and GCI Communication Corp)

1 Factual Information

1.1 Background

The 591-foot-long *Belle Mer* was a bulk carrier constructed of steel and built by Tsuneishi Group Shipbuilding in Zhoushan, China, in 2021. The vessel was owned by Reitaku Kaiun Co, Ltd and Belle Maritime S.A., and operated by Filteam Shipmanagement, Inc.

The Tongass Narrows is a body of water located between Gravina Island and Revillagigedo Island, Alaska, in the Clarence Strait. A subsea fiber optic cable, Alaska United Southeast Cable Segment 2 (AUSE-2), connected the cities of Ketchikan and Wrangell, Alaska, and lay along the bottom of the Tongass Narrows (see figure 2).² The cable was owned and maintained by GCI Communication Corp.

1.2 Event Sequence

On November 14, the *Belle Mer* was transiting southeast, bound for an anchorage site near Mud Bay in the Tongass Narrows, where it would onload a cargo of timber. While the *Belle Mer* was at anchorage, the logs would be floated from a nearby shoreside facility and pulled to the *Belle Mer* using assist boats.

About 0455, a Southeast Alaska Pilots' Association (SEAPA) pilot and two pilot observers (pilots in training) boarded the *Belle Mer* near the Guard Island Light in the Clarence Strait, 12 statute miles northwest of Ketchikan.³ The master and pilot conducted a master/pilot exchange, and the ship met an assist tug, which was then positioned on the *Belle Mer's* port bow to assist with anchoring.⁴

² The exact locations of the subsea cables are not readily available to mariners. Cable "as laid" surveys are maintained by subsea cable owners and were used to determine the cable's location.

³ A *pilot* is retained by the ship to provide local knowledge of the waterway, familiarity with tides and currents in the area, understanding of local procedures, and a thorough knowledge of the topography of the waterway. Pilots usually operate by issuing maneuvering instructions (such as heading, rudder angle, and speed orders) to the crew under the supervision of the master or the officer in charge of the navigation watch, or both.

⁴ A *master/pilot exchange* is required at the start of pilot transits and includes discussion of the vessel's navigational equipment, any limitations of maneuverability, available engine speeds, berthing maneuvers, intended course and speed through the waterway, anticipated hazards along the route, weather conditions, composition of the bridge team and deck crew both forward and aft including bow lookout, and so on.

The area chosen for the *Belle Mer* to anchor in the Tongass Narrows was not a common anchoring site. Most piloted traffic in the Tongass Narrows was cruise ships transiting through the area to dock in and near downtown Ketchikan. There were no US Coast Guard-designated anchorages near Mud Bay. The nearest designated anchorage, Ketchikan Harbor Anchorage Area, was 6 miles to the southeast, making it less desirable for the *Belle Mer*'s intended cargo operations.

The pilot, who had been a SEAPA pilot since 2007 and had experience anchoring cruise ships in the Tongass Narrows, told investigators that this was the first time he had anchored a vessel in this exact location. He recalled that smaller vessels, such as yachts, and at least one other logging vessel had been anchored near the *Belle Mer*'s location. The pilot stated that the stevedores and agents had requested the final anchoring location. Although nautical charts for the area outlined a cable area nearby where anchoring was restricted, he agreed to the requested location because it balanced the safety of the vessel with the requests of the stevedores and was outside of the anchoring-restricted area, all without impeding vessel traffic. The master, who had joined the *Belle Mer* a month prior, had never sailed in Alaska before and deferred to the pilot about the anchoring location.

While the pilot was conning the vessel to the anchoring location, the weather was calm. Winds were recorded at 5–10 knots, changing from the northwest to the southeast at Ketchikan International Airport (about 5 statute miles southeast of the anchoring location). The current in the area was slack, rising to 1.2 knots of flood at 0902. The pilot told investigators that he had checked the weather forecasts, noting increased winds expected for the next day. He shared this information with the master and offered guidance on preparing the vessel for the change in weather, telling him he could deploy 2 extra shots (about 180 feet) of anchor chain if needed.⁵ According to the pilot, this practice was "quite a normal procedure."

As the ship neared the anchoring location, the pilot instructed the crew to walk out 2 shots of anchor chain on deck and make the starboard anchor ready for letting go.⁶ (The *Belle Mer* was fitted with two stockless anchors, one each on the port and starboard sides of the bow.⁷ The starboard anchor was connected to 12 shots [about 1,080 feet] of anchor chain, and its hawsepipe was about 10 feet aft of the bow

⁵ A shot is a length of anchor chain equal to 15 fathoms (90 feet).

⁶ *Walking out* an anchor is the process of lowering it to the seafloor in a controlled manner, using an anchor windlass under power of its motor to control the anchor's descent.

⁷ A *stockless anchor* lacks a stock (a perpendicular crossbar at the top of the anchor shank) and is designed to lie flat on the seafloor.

stem.)⁸ The pilot communicated with a stevedore in a small boat nearby via handheld radio and light signals to help guide positioning. About 0550, the stevedore radioed to inform the pilot that the ship's position was acceptable. The pilot directed the ship to continue about a ship's length closer to shore, toward the southeast.

An astern engine order was given, and the AIS data shows that, about 0554, the *Belle Mer's* forward headway stopped. According to the master's statement, the order to let go the starboard anchor was given. When the *Belle Mer* began making sternway, the bow of the ship was about 190 feet east, and outside of, the charted cable and anchoring-restricted area. The water depth was about 220 feet. The pilot told investigators that the anchor was about 300 feet from the border of the restricted area.

The anchor chain was payed out in two-shot (180-foot) increments until 10 shots (about 900 feet) of anchor chain were on deck. As the *Belle Mer* backed and payed out the anchor chain, the pilot stated the ship took "a good strong strain on it," indicating to the pilot and master that the ship's anchor was holding in position. According to AIS data, the *Belle Mer's* astern momentum stopped about 0611. (The *Belle Mer's* GPS antenna, used to broadcast the vessel's position, was located was located 84.3 feet forward from the stern of the vessel, and 19.7 feet starboard of centerline.)⁹

The master told investigators that he and the third officer plotted a drag circle on the electronic chart display and information system (ECDIS) when the ship anchored.¹⁰ Although the pilot reported that the anchor was set on the seabed about 300 feet from the edge of the cable area, the ship's crew plotted the anchor point—the center of the drag circle—within the restricted area. The anchor point was also within a charted dumping ground.¹¹ The drag circle plotted by the crew had a radius

⁸ A *hawsepipe* is a cast-iron or steel pipe placed in the bow of a ship on each side of the stem for the anchor chains to pass through.

⁹ AIS equipment settings reflected a similar offset of the vessel's antenna, providing an accurate representation of the vessel's position on the ship's ECDIS.

¹⁰ (a) A *drag circle* is a circle centered on the anchor drop location and defined by the length of anchor chain that has been payed out plus the distance between the hawsepipe and GPS antenna. Drag circles are plotted so the crew can monitor whether the anchor is holding fast to the sea bottom. (b) The *Belle Mer* master was unable to explain to investigators how the crew plotted drag circles.

¹¹ A *dumping ground* (also known as a *spoil ground*) is an area of the seafloor designated for dumping of waste or dredged material. The water depth in the area and the seafloor's suitability for anchoring can be unpredictable due to the nature of the sediment or debris found there.

of 0.35 miles (2,127 feet) (see figure 5). About 0622, the pilot and pilot observers departed the ship via launch.

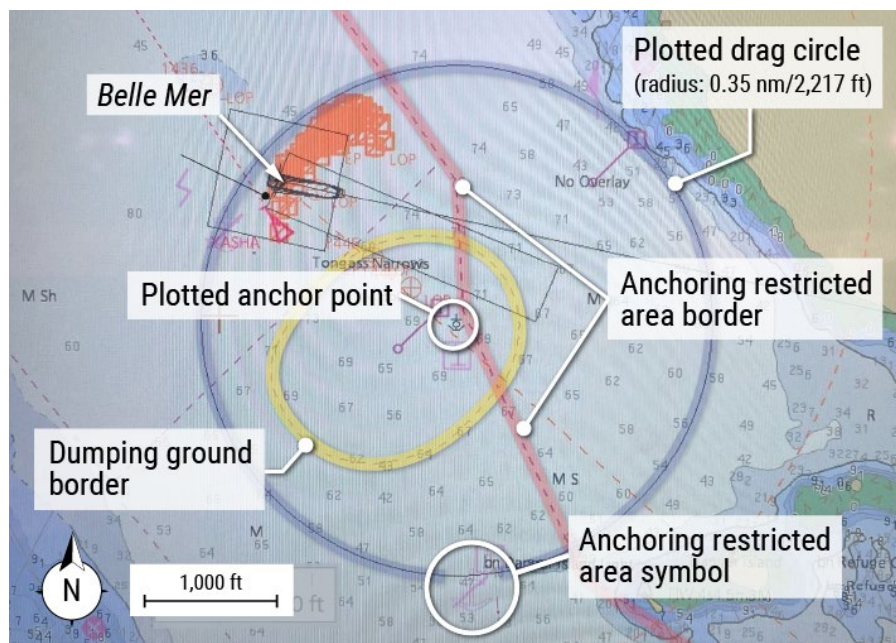


Figure 3. Photograph of the ECDIS screen, annotated to show the crew's plotted anchor point and drag circle (*highlighted in blue*) relative to the *Belle Mer*'s (*outlined in black*) movements as it swung at anchor (*orange*), the dumping ground (*highlighted in yellow*), and the anchoring restricted area's eastern border (*highlighted in red*). (Background source: Coast Guard)

Throughout the remainder of November 14 and into the next day, the second officer and third officer rotated anchor watches. The second officer stood watch from 1200 to 1800 and 0000 to 0600, with the third officer standing the opposite watches. Timber was floated from the shoreside facility near Mud Bay and pulled to the vessel, with stevedores unloading the timber cargo using the ship's cranes.

AIS data showed that, on November 15, at 0001, the *Belle Mer* was tending northwest from its original anchoring location. At this time, winds were less than 5 knots and generally from the northwest. As the ship tended toward the northwest, the distance between the ship's plotted positions and its original anchoring location gradually increased. Despite the ship's movement, its distance from the anchoring location never exceeded the 2,127-foot drag circle plotted by the master and third officer, and no additional chain was payed out during this period.

From 0300 to 1200, wind speeds increased from near calm to 15-20 knots, with gusts of 25 knots. Wind direction shifted from the northwest to the southeast. At 1200, winds were recorded at Ketchikan International Airport at 17 knots southeast, gusting to 26 knots. The master stated he observed 25-knot winds gusting to

33 knots. Additionally, the observed water levels at Ketchikan at this time were greater than 19 feet, about 15 feet higher than when the vessel originally anchored.

About 1211, the *Belle Mer* moved in a northwesterly direction, toward the cable. The vessel remained within the boundaries of the crew's plotted drag circle, so no ECDIS alarms were triggered (see figure 6). The master recalled that he and the crew were aware of the increased tension on the chain at this time, and that he believed the vessel's changing position was a result of the anchor chain "stretching."

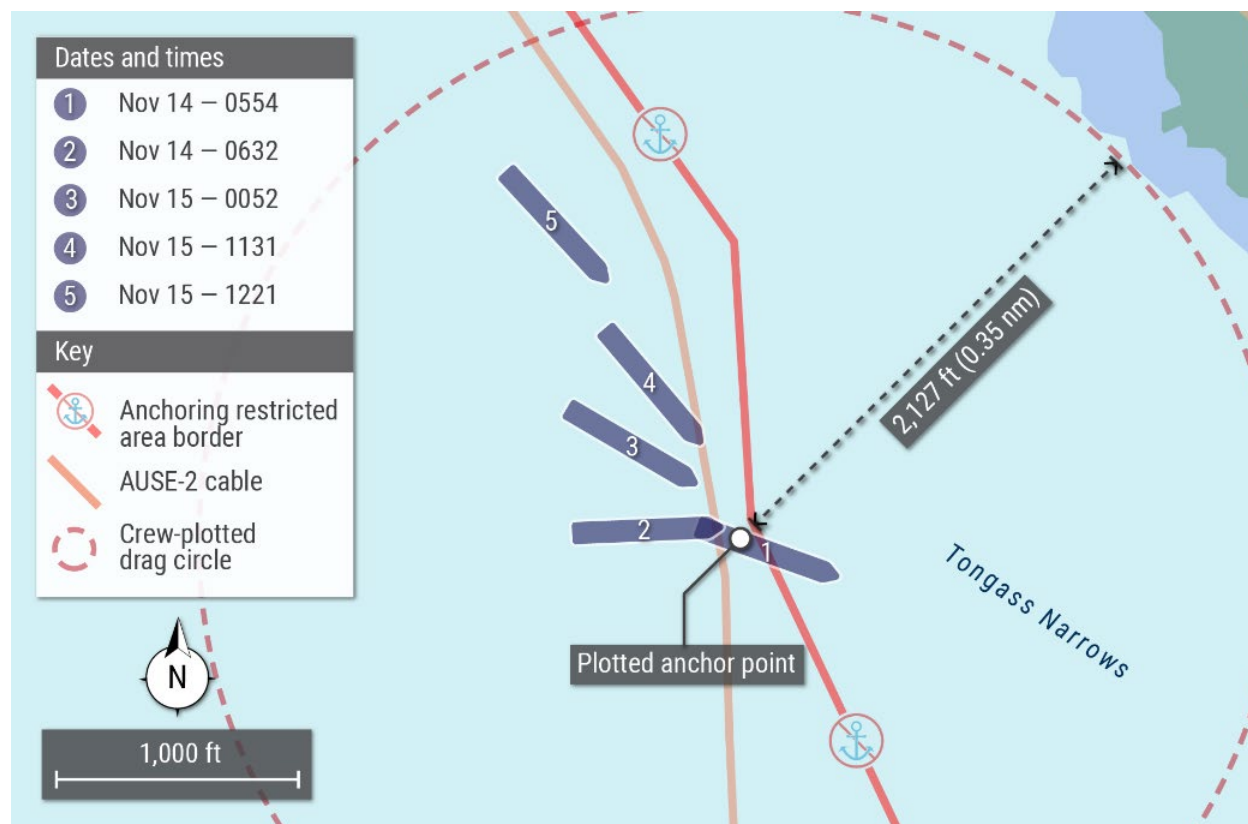


Figure 4. Selected AIS positions of the *Belle Mer* at anchor, showing the crew-plotted anchor point and drag circle radius relative to the approximate location of AUSE-2. (Background source: NOAA ENC US5KTNIC as viewed on Made Smart)

At 1226, the owner of AUSE-2 received an alert indicating a loss of data transmission on the fiber optic cable segment. The next day, November 16, about 0900, the *Belle Mer*'s agent notified the crew of an "underwater disturbance to an underwater cable" via email.

About 1300, cargo-loading operations concluded, and the *Belle Mer* remained at anchor while it awaited a cable repair ship to inspect the anchor and cable. On November 20, about 1245, the cable ship *Cable Innovator* arrived on scene and used a remotely operated vehicle (ROV) to confirm the anchor had become fouled on the

subsea cable (see figure 7). According to interviews, the ship's crew had not been aware the anchor had dragged. Later, the cable ship cut the cable and freed the anchor. That evening, about 2055, the *Belle Mer* departed for its next port of call in Prince Rupert, British Columbia, Canada.



Figure 5. Still from ROV footage showing the *Belle Mer*'s anchor fouled by cable wires. (Image brightness and contrast were adjusted for visibility.) (Background source: GCI)

1.3 Additional Information

1.3.1 Damage

The *Belle Mer* sustained no structural damage or damage to ground tackle equipment during the casualty. Nineteen of the 24 fibers in the AUSE-2 subsea fiber optic cable were damaged. GCI estimated repairs to the cable were \$240,000.

2 Analysis

On November 15, 2024, while the bulk carrier *Belle Mer* was anchored in the Tongass Narrows near Ketchikan, Alaska, the vessel began dragging anchor during a 3-hour period of sustained high winds, and about 1226, the vessel's anchor contacted AUSE-2, a subsea fiber optic cable.

2.1 Anchoring Location

On November 14, under the guidance of a SEAPA pilot, the *Belle Mer* approached the requested anchoring area in the Tongass Narrows, from the northeast. The pilot communicated with a stevedore in a nearby small boat, and the two agreed on an acceptable anchoring location. The master, who was unfamiliar with the Tongass Narrows, deferred to the pilot.

The ship was moved into position to begin anchoring. The *Belle Mer*'s engines were ordered astern and the forward headway stopped about 0554. The order to let go the starboard anchor was also given about this time. The AIS data and common anchoring practice indicate that the location of the starboard hawsepipe about 0554 is the likely location where the anchor initially rested on the seafloor.

The pilot reported he felt the chosen location was suitable and estimated the anchor was 300 feet from the border of a cable area (which marked the location of AUSE-2). Cable areas on nautical charts mark zones where one or more telecommunications or power cables are lying on, or buried in, the seafloor. Anchoring is restricted or prohibited in cable areas to protect undersea cables from damage by anchors. When the *Belle Mer*'s starboard anchor was let go, the bow of the ship was about 190 feet east, and outside of, the charted cable area—about 110 feet closer to the cable area than the pilot had estimated. Based on AIS data and “as laid” survey data, the NTSB estimates the location where the anchor was dropped and initially rested was as close as 445 feet from AUSE-2.

In addition, the anchor's location fell within a charted dumping ground (see figure 5). Although anchoring in a dumping ground is not prohibited, it is inadvisable due to the unpredictable depth and composition of the seafloor in such an area. When agreeing to an anchoring location, the pilot and master should have considered the distance to the charted hazard (subsea cable) and the suitability of the seafloor for anchoring, accounting for possible drag.

Ten shots of chain were deployed while the *Belle Mer* moved astern. When the *Belle Mer*'s astern momentum stopped about 0611, the pilot reported that the ship took a “good, strong strain” on the chain.

2.2 Anchor Chain Scope

In addition to proximity to hazards and suitability of the seafloor, pilots and masters should consider the water depth, the length of available anchor chain, and forecasted weather conditions when they plan to anchor. Maintaining adequate scope—the ratio of the length of anchor chain deployed from the ship to the depth of the water under the bow of the vessel—ensures that an anchor’s designed holding power is achieved and prevents dragging.¹² Deploying a length of anchor chain with a scope of 5:1 to 7:1 is recommended; even more anchor chain is recommended in adverse weather.¹³

The ship anchored at a depth of 220 feet. The crew deployed 900 feet (10 shots) of the available 1,080 feet (12 shots) of the starboard anchor chain; the deployed chain’s scope was about 4:1. To anchor a vessel in moderate conditions, a scope of 5:1 (1,100 feet; 12.2 shots) to 7:1 (1,540 feet, 17 shots) is recommended. If the crew had deployed the remaining 2 shots, the chain’s scope would have been closer (4.9:1) to the minimum but still under the lower end of the recommended range. Additionally, given the forecasted adverse weather (increased wind speeds), more scope would have been prudent. Therefore, given the water depth, the starboard anchor chain’s scope was inadequate for both moderate conditions and the high winds on the day of the casualty.

The day after the *Belle Mer* anchored, wind speeds and the tide were higher than the previous day, and the vessel’s AIS positions relative to its anchor’s location reflected a vessel whose anchor chain was under tension. Given the chain’s inadequate scope, it is likely the anchor chain had minimal catenary. When a chain has minimal catenary, the holding power of the anchoring system (chain and anchor) is negatively affected. The chain’s frictional holding power is nearly eliminated (as the chain is lifted from the seafloor) and the reduced angle at which the anchor chain pulls on the anchor decreases the effective holding power of the anchor. Diminished holding power, in turn, increases the likelihood of the anchor unseating and dragging. Due to anchor chain scope, the *Belle Mer* likely dragged anchor into the nearby restricted area until the anchor struck AUSE-2.

¹² Scope is the ratio of the length of anchor chain paid out to the depth of water.

¹³ Nathaniel Bowditch, *American Practical Navigator: An Epitome of Navigation*, vol. 1, 2024 ed. (US National Geospatial-Intelligence Agency, 2024), 212–213, https://msi.nga.mil/api/publications/download?key=16693975/SFH00000/Bowditch_Vol_1.pdf.

2.3 Drag Circle

Once the vessel was anchored, the crew plotted a drag circle on the vessel's ECDIS. According to the *Belle Mer*'s master and ECDIS photos, the drag circle was plotted with a 2,127-foot (0.35-mile) radius. A drag circle radius is calculated using the following formula:

$$D\pi = B + (S \times 90 \text{ ft})$$

Where:

$D\pi$ = drag circle radius

B = distance between the hawsepipe and GPS antenna on the vessel
(in plain view/longitudinally)

S = number of shots payed out.¹⁴

Given the distance from the *Belle Mer*'s hawsepipe to its GPS antenna (495 feet) and the amount of anchor chain payed out (10 shots; 900 feet total), the drag circle calculations should have yielded a 1,395-foot radius rather than the 2,127 feet (0.35 miles) plotted by the crew (see figure 8). In addition, the center of a drag circle should be plotted using the location of the anchor drop. However, the crew plotted the drag circle's center 465 feet northwest of where the NTSB estimated the anchor was dropped about 0554. Investigators noted that the plotted location corresponded with the location of the ship's GPS antenna at 0554. Therefore, it is possible the crew incorrectly used the location of the GPS antenna rather than the location where the anchor was dropped.

¹⁴ Bowditch, 212.

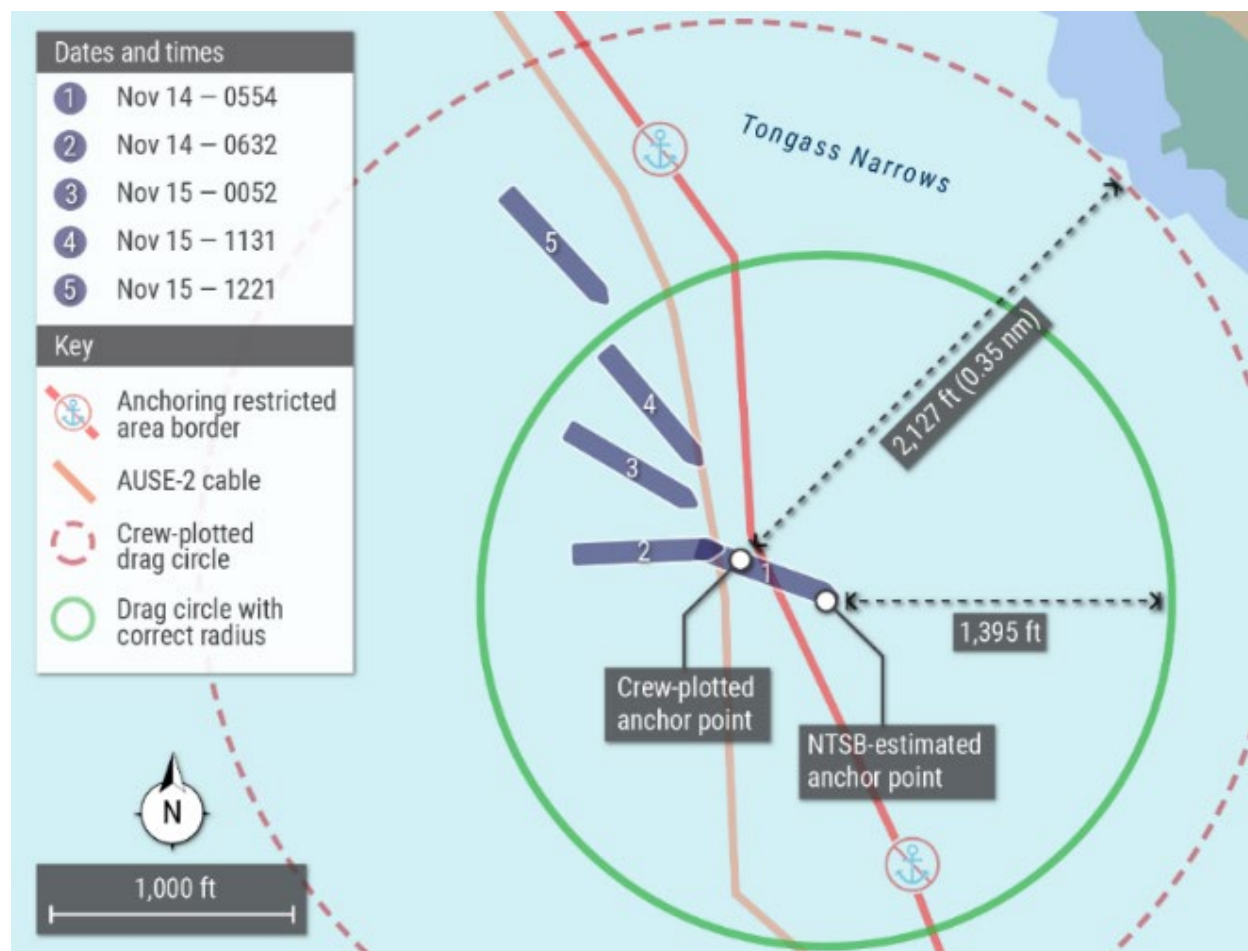


Figure 6. Selected AIS positions of the *Belle Mer* at anchor, showing the approximate location of AUSE-2, and a comparison of the crew-plotted anchor point and drag circle radius with the drag circle with the correct radius based off the NTSB-estimated anchor point. (Background source: NOAA ENC US5KTNIC as viewed on Made Smart)

Investigators also noted that the radius of the crew's plotted drag circle matched the distance to the nearest shoal water—0.35 miles—from that plotted anchor location. Shoal waters lined the coast near the anchoring location the stevedore requested and were visible on the vessel's ECDIS display, and the crew would have known to avoid the areas (see figure 9). It is possible that the borders of the plotted drag circle were based on proximity to shoal water. However, the master was unable to explain the crew's process for plotting drag circles, and no other crewmembers were interviewed. Therefore, the exact method the *Belle Mer* crew used to plot the vessel's drag circle could not be determined.

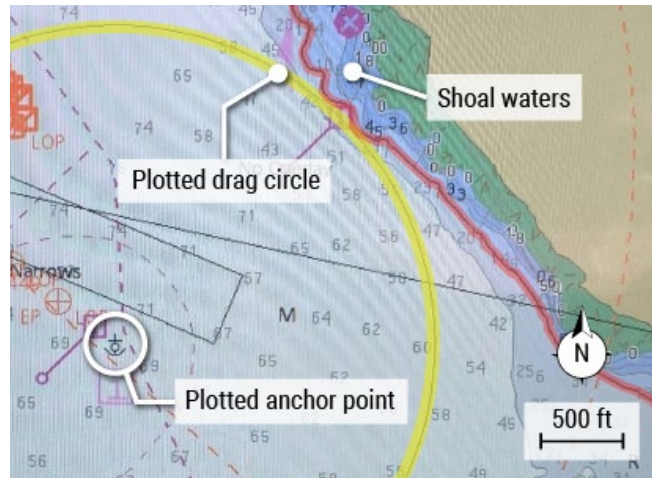


Figure 7. The crew's plotted drag circle (*highlighted in yellow*) relative to nearby shoal waters (*border highlighted in red*). (Background source: Coast Guard)

On November 15, about 1211, the vessel was 1,395 feet from the actual anchoring location. For the next 15 minutes, the vessel moved in a northwesterly direction, past the cable. No ECDIS alarms alerted the crew during this time because the vessel never exceeded the bounds of the drag circle they plotted. The crew's drag circle, which was not based on the length of anchor chain paid out or the distance from the hawsepipe to the ship's GPS antenna, was too large to effectively evaluate whether the ship was dragging anchor. If the crew had correctly plotted the drag circle radius, an ECDIS alarm may have notified the crew as the vessel moved past the circle's boundary. Without an audible alarm, the crew remained unaware that they were dragging anchor. Had the crew properly plotted the drag circle correctly, they may have had the opportunity to take action to attempt to avoid the anchor striking the cable.

3 Conclusions

3.1 Probable Cause

The National Transportation Safety Board determines that the probable cause of the anchor strike of bulk carrier *Belle Mer* with the Alaska United Southeast Cable Segment 2 subsea fiber optic cable was the pilot and master's choice to anchor near a known restricted anchoring area with inadequate anchor chain scope given the water depth and forecasted weather conditions. Contributing to the casualty was the *Belle Mer* crew's improperly plotted drag circle, which was too large and, therefore, could not be used to effectively evaluate whether the ship was dragging anchor.

3.2 Lessons Learned

Anchoring Considerations

When choosing a suitable anchoring location, mariners must consider many different criteria, including water depth, bottom type, proximity to hazards, weather, and available anchor chain scope. Holding power of the anchor and anchor chain also should be constantly analyzed while at anchor, especially under increasing external forces (such as wind and current) and while anchored near charted hazards. Additionally, drag and swing circles should be based on the actual anchor drop location and accurate vessel and anchor chain measurements. Once accurate drag and swing circles are plotted, mariners should continually monitor the vessel's position with regular fixes compared to the plotted drag circle, swing circle, and available visual references.

Vessel Particulars

Vessel	<i>Belle Mer</i>
NTSB vessel group	Cargo, Dry Bulk (Dry bulk carrier)
Owner/operator	Reitaku Kaiun Co., Ltd. (Owner) Filteam Shipmanagement, Inc. (Operator)
Flag	Panama
Port of registry	Panama
Year built	2021
Official number	N/A
IMO number	9897951
Classification society	Nippon Kaiji Kyokai (ClassNK)
Length (overall)	590.5 ft (180.0 m)
Breadth (max.)	105.0 ft (32.0 m)
Draft (casualty)	30.0 ft (9.1 m)
Tonnage	26,269 GT ITC
Engine power; manufacturer	8,595 hp (6,409 kW); Mitsubishi diesel engine

NTSB investigators worked closely with our counterparts from **Coast Guard Marine Safety Detachment Ketchikan** throughout this investigation.

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For more detailed background information on this report, visit the [NTSB Case Analysis and Reporting Online \(CAROL\) website](#) and search for NTSB accident ID DCA25FM005. Recent publications are available in their entirety on the [NTSB website](#). Other information about available publications also may be obtained from the website or by contacting—

National Transportation Safety Board
Records Management Division, CIO-40
490 L’Enfant Plaza, SW
Washington, DC 20594
(800) 877-6799 or (202) 314-6551